

Advanced Robotics - HW4 Writeup

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Bayes Filter

The predict function was written to update the state belief array by iterating through each cell and finding the new probability of being in that cell and adjacent cells based on the last probability of being in that cell. Probabilities are then normalized to sum to 0.

The update function was written to then update the state belief for each cell in the array by comparing expected sensor measurements (for each heading) with actual sensor measurements.

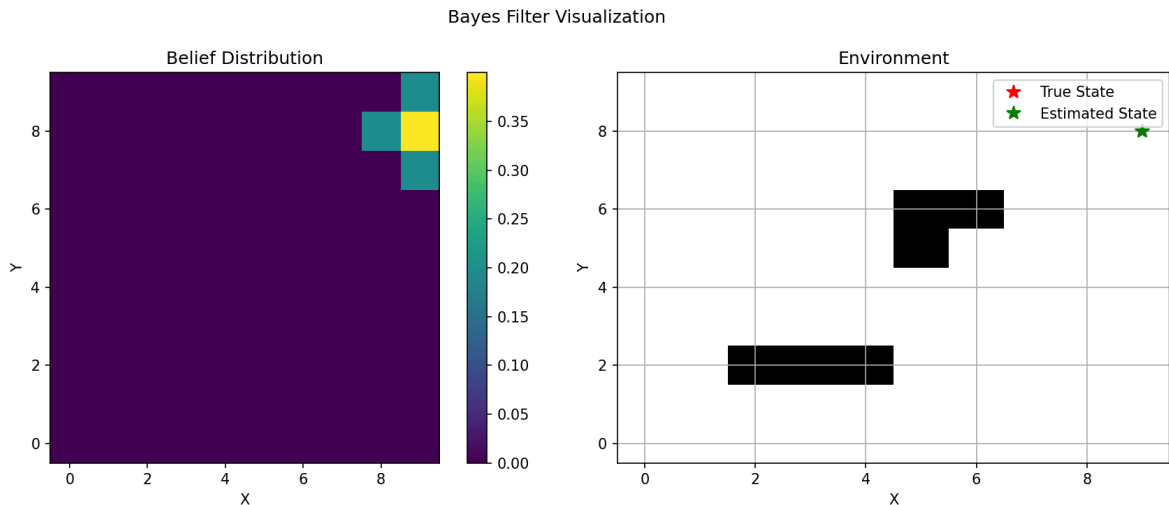


Figure 1 Bayes filter final state estimate

Extended Kalman Filter

The motion model function was written to estimate the new state of the robot by propagating the motion model. Since velocity is scaled based on a function of position, the new state was determined by integrating the product of the velocity inputs and the velocity scaling at that point over the timestep.

The motion Jacobian function was written to compute the Jacobian of the nonlinear motion model with respect to the state vector. To do so, partial differentiation was performed on each dynamics equation with respect to each state variable.

The measurement Jacobian function computes the error between real and expected measurements, and the partial derivatives of the range (hypotenuse) of the predicted range with respect to each state variable.

The predict function was written to update the new state and covariance using the motion model and motion Jacobian functions respectively, and the update function updates the state and covariance based on the measurement model and measurement Jacobian functions.

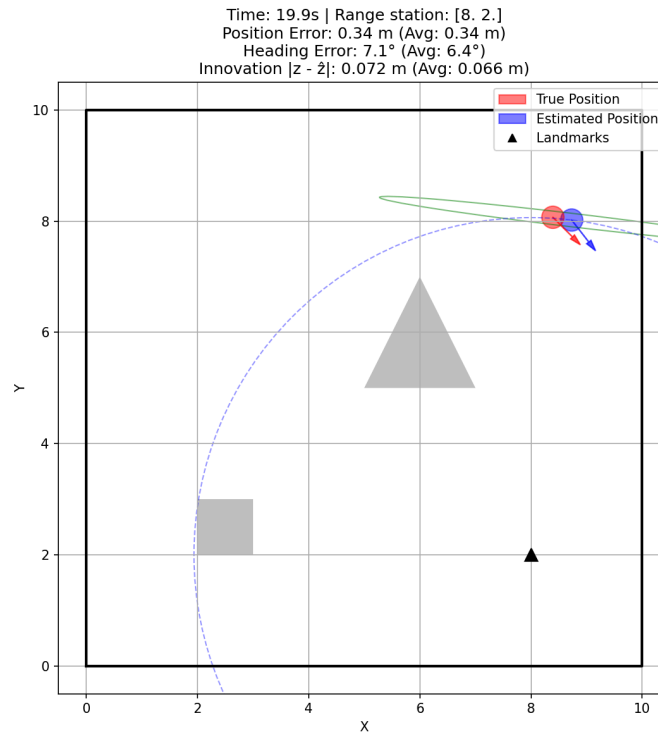


Figure 2 EKF final state estimate

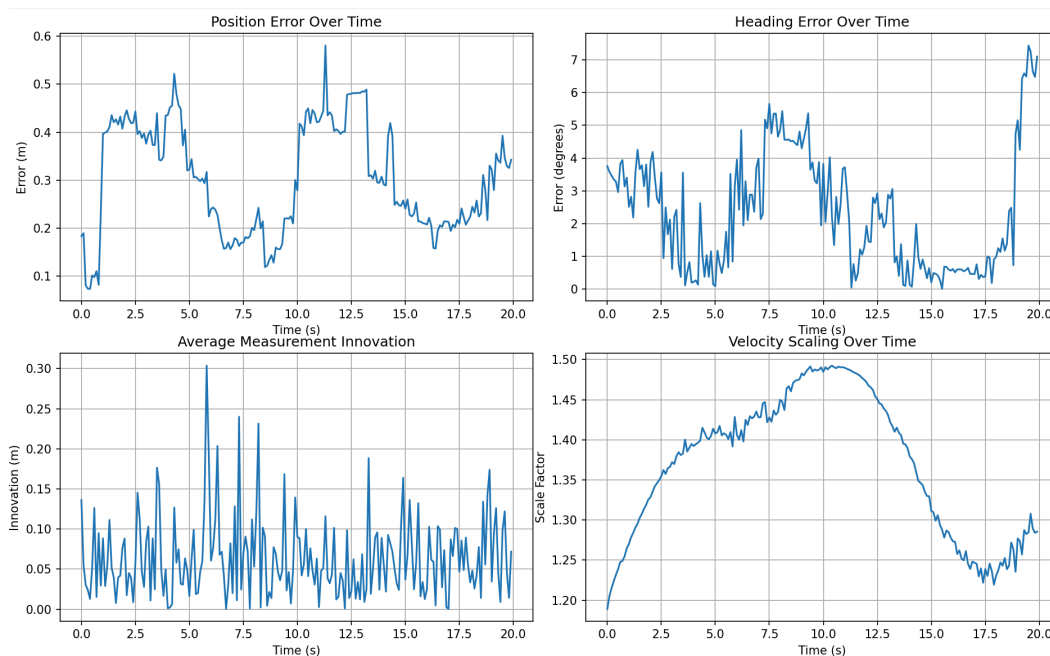


Figure 3 EKF performance graphs

Particle Filter

The predict function propagates each existing particle through the motion model based on control inputs with noise by integrating the x, y and angular components of the noisy velocity inputs over the timestep. For any particle that moves out of bounds or into an obstacle, it is resampled at a random point in the environment.

The update function takes these new particles and updates their importance or accuracy weights based on how well the expected measurements of their respective states match the actual measurements of the robot. These weights are then normalized to add up to one.

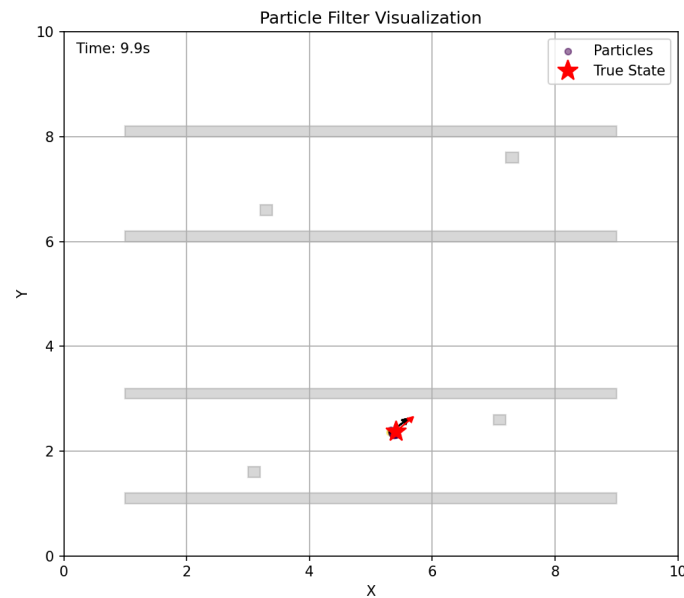


Figure 4 Particle filter final state estimate